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REPETITIVE PEER SEPARATION OF YOUNG MONKEYS: EFFECTS OF VERTICAL CHAMBER CONFINEMENT DURING SEPARATIONS [1]

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Four rhesus monkeys were reared together, without mothers, for the first 3 months of life, then subjected to 20 peer separations and reunions during the next 6 months, being confined to vertical chambers during the periods of separation. Their behaviors were compared with those of four monkeys who were reared together without separations and four monkeys from a previous study who had been subjected to similar separations but not confined to vertical chambers. It was found that both groups of separated monkeys exhibited protest–despair reactions to the separations and in comparison with the nonseparated controls showed maturational arrest. However, the chamber-separated subjects exhibited higher levels of self-directed activity and lower levels of socially directed activity than their cage-separated counterparts. The significance of these findings with respect to monkey models of human anaclitic depression is discussed.

Experimentation designed to simulate in nonhuman subjects behavioral syndromes akin to those exhibited by humans possessing affective disorders has been pursued for nearly half a century (Liddell, James, & Anderson, 1934; Masserman, 1943; Pavlov, 1927). A recent emphasis has focused on production of depression or depressivelike behaviors in young monkeys. The initial basis for this work has come from clinical observations of young children who experienced maternal separation via hospitalization. Both Spitz (1946) and Bowlby (1960, 1969) reported that some children separated from their mothers for a prolonged time developed a syndrome characterized by initial agitation ("protest") followed by extended withdrawal and inactivity ("despair"), a reaction Spitz termed "anaclitic depression." Whereas Spitz observed virtually immediate and complete recovery in his children, 6–12 months of age, upon reunion with their mothers, Bowlby reported indifference and/or hostility ("detachment") following maternal reunion among some of his children who were 3–5 years old.

Researchers at the Wisconsin Primate Laboratory produced a similar reaction to maternal separation in young rhesus monkeys (Seay, Hansen, & Harlow, 1962; Seay & Harlow, 1965). Infant monkeys were reared with their mothers and permitted peer interaction for approximately the first 6 months of life, separated from their mothers for a period of 2 or 3 weeks, then reunited with their mothers. Like the children observed by Spitz and Bowlby, these monkeys exhibited a protest–despair reaction to maternal separation, with protest characterized by high activity and vocalization and despair characterized by inactivity, social withdrawal, and lack of social play. Mother–infant ties were quickly reestablished following reunion, providing little evidence for the occurrence of Bowlby's detachment phenomenon in the young monkey subjects. Numerous investigators working at other laboratories and employing a variety of macaque species have reported similar reactions of infants to maternal separation (Hinde & Spencer-Booth, 1971; Hinde, Spencer-Booth, & Bruce, 1966; Jensen & Toleman, 1962; Kaufman & Rosenblum, 1967a), and

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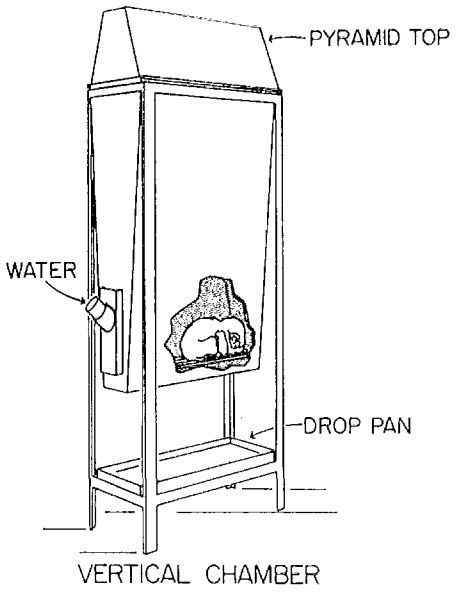


FIG. 1. The vertical chamber apparatus.

the mother–infant separation paradigm has been posited as a basis for development of a monkey model of human anaclitic depression (Harlow, Suomi, & McKinney, 1970; Kaufman & Rosenblum, 1967b; McKinney & Bunney, 1969).

Subsequent researchers have indicated that depressivelike behaviors can be elicited in young rhesus monkeys without recourse to maternal separation. Suomi, Harlow, and Domek (1970) reared a group of infants together for the first 3 months of life without mothers. During the next 6 months the subjects were separated from each other, then reunited, a total of 20 times, with each separation being 4 days in duration. It was found that the subjects exhibited a protest–despair reaction to each of the 20 separations. Immediately following each separation the infant monkeys showed high levels of vocalization and activity. Subsequently, in each separation period, these behaviors diminished and were replaced by high levels of self-clasping and huddling behaviors. Each reunion was characterized by high levels of mutual clinging behaviors by the subjects. Moreover, the subjects did not adapt to the multiple-separation procedure. Their reactions to the final separations were essentially as pronounced as their reactions to the initial separations. Finally, a cumulative result of the multiple separations was an arrest of normal maturation of behaviors in the subjects. Their behavior repertoire at age 9 months, following the separations, was virtually identical to their behavior repertoire at age 3 months, prior to the separations, in sharp contrast to the extensive behavioral development commonly exhibited by rhesus monkeys during this chronological period. Suomi et al. thus demonstrated that multiple peer separations consistently produced behaviors comparable to those resulting from mother–infant separations. Unfortunately, the lack of a group of nonseparated control subjects weakened the design of the study, although approximate control data were available from previous studies.

An alternative approach to the production of depressivelike behaviors in young monkeys has utilized the Harlow vertical chamber apparatus (Suomi & Harlow, 1969), basically a stainless steel trough with sides sloping inward to a rounded bottom (see Figure 1). Depression in humans has been described as "embodying feelings of helplessness and hopelessness, sunken in a well of despair [Schmale, 1970, p. 1]," and the chamber was intuitively designed to reproduce such a well physically and psychologically for monkey subjects.

Two researchers indicated that vertical chamber confinement for relatively brief time periods resulted in prolonged psychopathological behavior of a depressive nature for young socially unsophisticated monkeys. Harlow and Suomi (1971) incarcerated four wire-cage-reared subjects 6–13 months of age for a total of 30 days each. These monkeys displayed significant increases in self-clasping and huddling behaviors and significant decreases in locomotor and exploratory behaviors following chamber confinement, in comparison with their prechamber base lines and with a nonchambered control group. Suomi and Harlow (1972) confined four subjects 45–90 days of age in vertical chambers, then tested the subjects with both socially sophisticated and socially unsophisticated control subjects until 1 year of age. The chambered subjects exhibited significantly higher levels of self-clasping and huddling behaviors and significantly lower levels of locomotion, ex-

ploration, social contact, and play than the controls during the entire period of testing.

These findings were of interest in two respects. First, although vertical chamber confinement clearly represented a form of social isolation, it yielded subject disturbance beyond that achieved via traditional isolation procedures. Previous research had indicated that neither isolation of less than 6 months in duration (Boelkins, 1963; Griffin & Harlow, 1966) nor isolation not initiated at birth (Clark, 1968; Rowland, 1964) resulted in severe and persistent psychopathology. In contrast, vertical chamber confinement of only 30 days in duration initiated as late as 13 months of age had profound and prolonged consequences. Second, vertical chamber confinement of the above subjects did not represent a social manipulation—at least not one of breaking existing social attachment bonds —because the monkeys did not have the opportunity to form any social attachments with conspecifics prior to incarceration. Nevertheless, their behaviors following vertical chamber confinement were virtually identical to those exhibited by monkeys during the despair stage of reaction to social separation, clearly a social manipulation.

The following study represents an attempt to combine the manipulations of vertical chamber confinement and repetitive peer separation. Specifically, infant monkeys were reared together, without mothers, for the first 3 months of life, then subjected to a total of 20 separations and reunions according to the same schedule employed in the Suomi et al. (1970) study. However, the subjects were confined to vertical chambers during the separations instead of to individual cages as in the previous study. In addition, a group of nonseparated subjects was also studied in order to provide the control data which the Suomi et al. study lacked.

METHOD

Subjects

Chamber-separated subjects. The chamber-separated subjects were four rhesus monkeys, two males and two females, born within a 9-day span. They were separated from their mothers at birth and reared in the laboratory nursery according to the procedure described by Blomquist and Harlow (1961). When their mean age was 15 days, the animals were placed as a group of four in a $6 \times 3 \times 4$ ft. cage, where they remained undisturbed until their mean age was 90 days.

Control subjects. The control subjects were two male and two female rhesus monkeys, born within 10 days of each other. They were separated from their mothers at birth and reared in an identical fashion as the chamber-separated subjects for their first 90 days of life.

Procedure

The procedure used for the chamber-separated monkeys was virtually identical to that of Suomi et al. (1970). Beginning at 90 days of age, the monkeys were subjected to a series of 20 separations and reunions; each separation was 4 days in duration and each reunion was 3 days in duration. During the reunions the subjects were housed in their group cage. Between the twelfth and thirteenth separation was a 6-week period during which the subjects remained in their group cage as a group of four. Following the twentieth separation and reunion the subjects continued to be housed in their group cage for an additional 6 weeks. However, when separated, the subjects were not individually housed in single $2.5 \times 2 \times 2.5$ ft. cages as in Suomi et al. Instead, they were placed in individual vertical chambers, where they could hear, but neither see nor touch each other. Thus, the experimental subjects in this study were subjected to 20 4-day periods of separation during which they were confined to vertical chambers. For the remainder of their first 10 months of life they were housed as a group of four. Control subjects remained undisturbed as a group of four in their group living cage throughout the $10\frac{1}{2}$ months of the experiment.

Data Collection

The two observers who scored the monkeys' behaviors for Suomi et al. (1970) collected the data for this study. Chamber-confined subjects were observed once daily, beginning 6 weeks prior to the first separation and continuing for 6 weeks after the final reunion, except for the periods during which they were confined to the vertical chambers. The decision to omit vertical chamber observations was based on two considerations. First, the chambers were fitted with pyramidal tops covered by fine mesh, making accurate recording of subject behavior within the chambers difficult. Second, it was feared that positioning an observer above the chamber would tend to agitate the subject, and since the purpose of the experiment included production of depressive behavior, avoidance of any possible source of agitation of subjects when in the chambers was desired. Control subjects were observed once per day, 5 days per week, for the duration of the experiment.

As in Suomi et al. (1970), observation sessions of the subjects in the group cage lasted 30 minutes, during which time each group member was scored for 7.5 minutes. Measures consisted of frequency and

duration during each 7.5-minute period of subject behaviors falling into each of 10 behavioral categories. The categories utilized were identical in definition to those employed in Suomi et al., and their descriptions are as follows:

Partner cling—contact of own ventral body surface with another subject's.
Self-clasp—clasping of any part of own body with hand(s) and/or foot (feet).
Self-mouth—oral contact to own body.
Rock and huddle—self-clasping accompanied by stereotypic rocking.
Passive—absence of all social, exploratory, and locomotor activity.
Locomotion—any self-induced change in physical location of self.
Environmental exploration—tactual and/or oral manipulation by subject of inanimate objects, for example, towel, toy, cage lock, etc.
Social contact—tactual and/or oral contact with another subject other than ventral clinging.
Vocalization—any sound emitted by the subject.
Play—any type of socially directed play activity (including rough and tumble, approach–withdraw, and noncontact-type play).

Order of testing among subjects was randomly determined. Observations were made between the hours of 9:00 A.M. and 4:30 P.M.

Data Analysis

Because subjects in the present experiment were reared and observed in group cages of identical size by the same observers during the precise chronological period that the subjects in Suomi et al. were reared and observed, data from that study were included in the present analyses. Thus, a direct comparison between the monkeys in the present experiment and those subjected to 20 repetitive infant–infant cage separations was possible. The analyses were carried out in three stages.

First, behaviors of the chambered subjects in the present experiment and those of the subjects in Suomi et al. were compared over the 12 72-hour reunion periods of the first separation series. Separate analyses were performed for the categories of self-mouth, self-clasp, locomotion, environmental exploration, and partner cling. For each category, a mean was calculated for each subject over each of three time blocks: Reunions 1–4, Reunions 5–8, and Reunions 9–12. Each mean represented the average number of seconds per 7.5-minute observation session that a subject was observed performing behaviors encompassed by the category during the time block. The means were then subjected to a two-way repeated-measures analysis of variance, with group of subject (chamber separated *versus* cage separated) as the independent variable and time block as the repeated measure. For those categories whose analysis had yielded a significant time block effect and/or a significant interaction, a Duncan new multiple-range test (Duncan, 1955) was subsequently performed.

Similar analyses were carried out on the scores obtained for both groups of subjects during the eight reunions of the second separation series. A sixth behavioral category, play, was added for analysis. Two levels of the time block variable, one encompassing Reunions 13–16 and the other encompassing Reunions 17–20, were examined in the analyses. In all other respects, the form of analysis was identical to that used for the first series of reunions.

Finally, behavioral levels of all subjects during the three base-line periods were directly compared. Six behavioral categories—self-mouth, self-clasp, locomotion, environmental exploration, partner cling, and play—were analyzed separately. For each category, a mean for each subject during every base-line period was calculated, representing the average number of seconds per 7.5-minute observation session during which subject behaviors encompassed by the category were observed. The means were then subjected to a two-way repeated-measures analysis of variance, with group of subject (chamber separated vs. cage separated vs. control) as the uncorrelated variable and base-line period (3, 6, or 9 months of age) as the repeated measure. A Duncan test was performed on the means of those categories whose analysis had yielded any significant main effects and/or a significant interaction.

RESULTS

The results indicated that the monkeys confined to vertical chambers during the repetitive infant–infant separations exhibited protest–despair reactions to the separations and in comparison with nonseparated controls showed severe maturational arrest over the course of the experiment. However, their reactions during the reunion periods and later base-line periods were of a significantly different form than those exhibited by the monkeys in the Suomi et al. (1970) study which were subjected to an identical separation procedure but were confined to cages, rather than vertical chambers, during their terms of separation.

Because no formal observations were made of the chambered subjects during the periods of separation, quantitative assessment of their reactions to the repetitive separations was not possible. However, anecdotal observations made by both the investigators and caretaking personnel over the course of the study strongly suggested that these monkeys exhibited a biphasic reaction to each separation, characterized by initial agitation followed by a period of inactivity. In particular, the subjects vocalized at exceedingly high levels and moved about inside the vertical chambers for

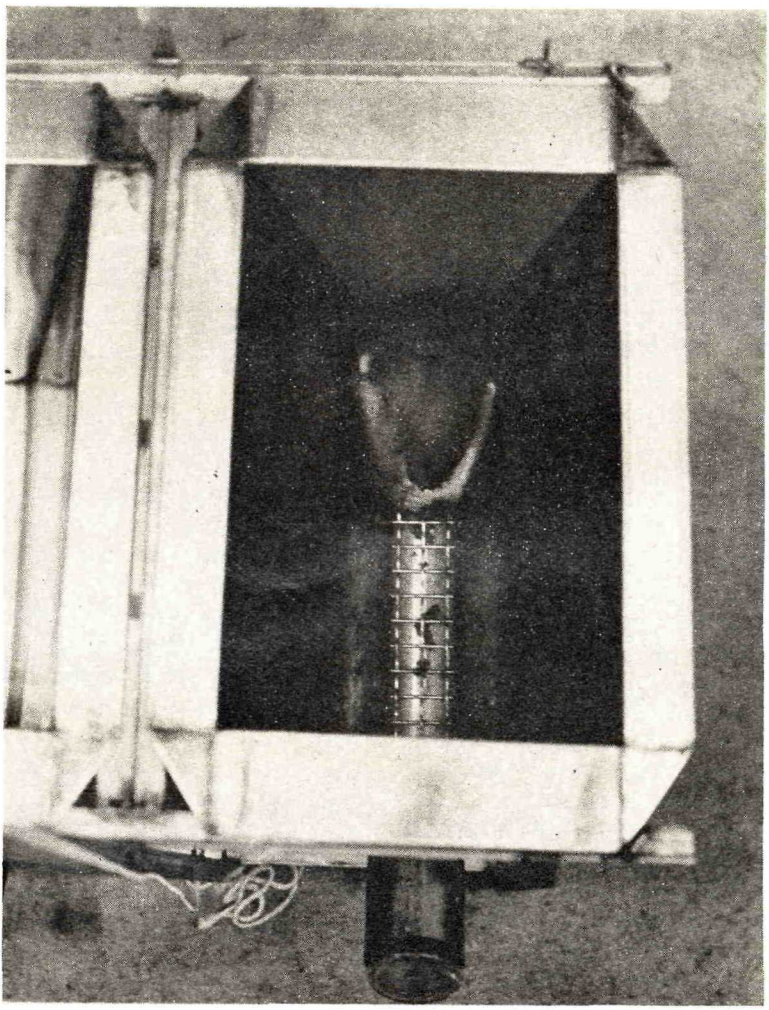


FIG. 2. Typical subject posture within the vertical chamber.

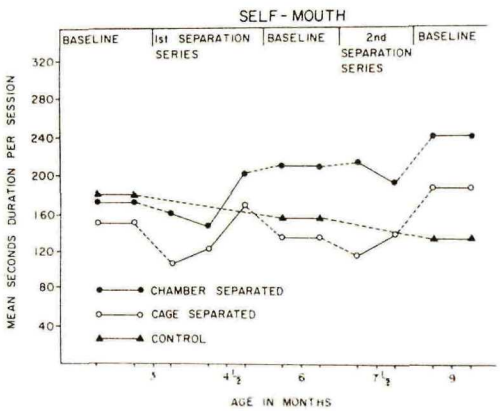


FIG. 3. Mean levels of self-mouth.

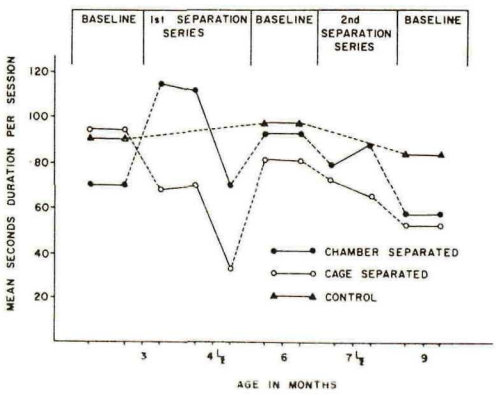


FIG. 4. Mean levels of environmental exploration.

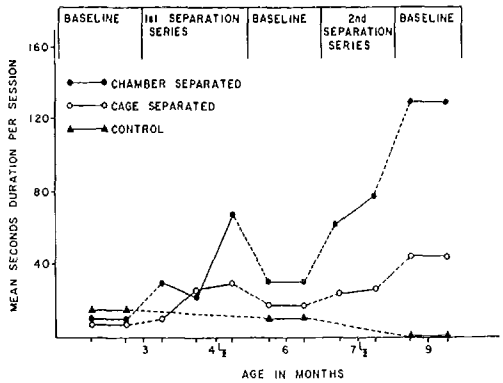


FIG. 5. Mean levels of self-clasp.

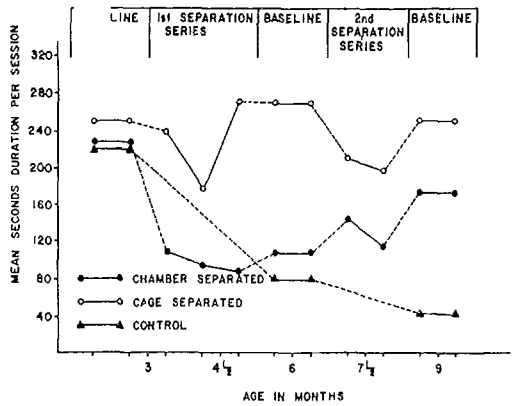


FIG. 7. Mean levels of partner cling.

approximately the first day of each separation. By the third or fourth day of the separations, however, the monkeys typically had assumed a huddling posture in a corner of the chamber (see Figure 2). Thus although quantitative data were lacking, there existed strong suggestive evidence that these subjects underwent successive protest–despair reactions to the repetitive separations.

Quantitative data were obtained from the formal observations taken during the baseline and reunion periods. Behavioral levels derived from these observations for the chamber-separated and cage-separated subjects during the reunion periods of both separation series, as well as the three base-line levels for all subjects, are plotted in Figures 3–8.

The analyses of reunion behaviors for the first separation series are summarized in Table 1. Chamber-separated and cage-separated subjects exhibited similar trends for the behaviors of self-mouth, environmental exploration, and locomotion, although the cage-separated subjects had significantly lower levels of the latter behavior than the chamber-separated subjects. However, the groups differed markedly with respect to the behaviors of self-clasp and partner cling. Chamber-separated monkeys exhibited higher levels of self-clasping, particularly during the final four reunions of the series. In contrast, cage-separated subjects exhibited consistently higher levels of partner cling than the chamber-separated subjects, with the differences most marked during the final four reunions. Analyses of reunion behaviors for the second separation series (Table 2) indicated similar reactions to reunion among both groups, save for consistently higher levels of self-mouthing and self-clasping for chamber-separated subjects.

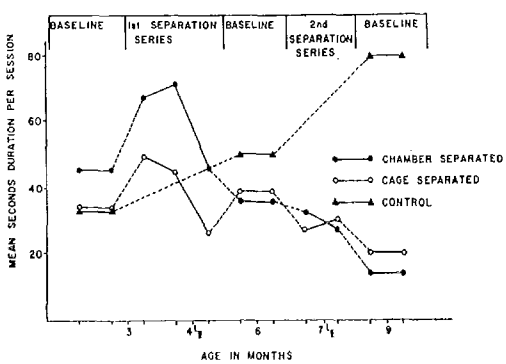


FIG. 6. Mean levels of locomotion.

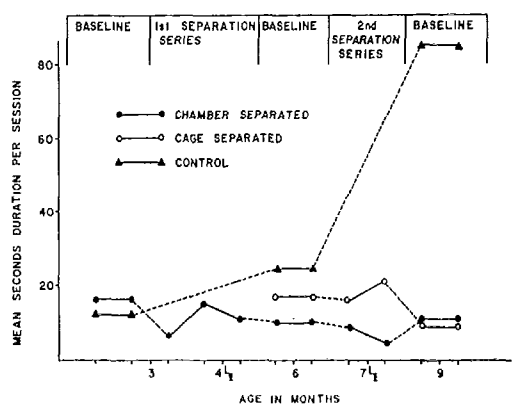


FIG. 8. Mean levels of play.

TABLE 1

RESULTS OF ANOVA AND DUNCAN TESTS, FIRST SEPARATION SERIES (SEPARATIONS 1–12)

Category	Group (G)[a]	Time block (T)[b]	G × T[b]	Duncan test
Self-mouth	1.91	7.97**	<1	3 > 1,2**
Environmental exploration	5.40	9.47**	<1	3 < 1,2**
Self-clasp	6.06*	14.43***	6.27**	VS3, VS1, VS2, CS1, CS2, CS3** CS1 < VS1, CS3*
Locomotion	19.99**	22.13***	1.17	3 < 1,2**
Partner cling	14.97**	9.93**	11.67***	VS3, VS2, VS1 < CS3, CS2, CS1** CS2 < CS1, CS3**

Note. Tabled numbers are *F* scores. Abbreviations for Duncan tests: VS = vertical chamber separated, CS = cage separated; 1 = Reunions 1–4, 2 = Reunions 5–8, 3 = Reunions 9–12.

[a] $df = 1/6$.
[b] $df = 2/12$.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

The cumulative effects of the multiple separations were most evident in the analyses of behavioral levels of all subjects during the three base-line periods (Table 3). Prior to the separations all subjects exhibited similar levels of all behaviors scored. But during the latter two base-line periods maturational arrest was clearly evident among the cage- and chamber-separated subjects. Both groups failed to show the subsequent increases in locomotion and play and the sharp decrease in partner cling displayed by the nonseparated control monkeys. Moreover, the form of the maturational arrest differed significantly between the two separation groups. Chamber-separated subjects exhibited significantly higher levels of self-mouth and self-clasp and significantly lower levels of partner cling than their cage-separated counterparts.

DISCUSSION

The results of this study indicated that monkeys repetitively separated from each other not only displayed protest–despair reactions to the separations but also failed to exhibit behavioral development comparable to that of monkeys reared together from birth but not subjected to repetitive separations. The nonseparated control monkeys showed progressive increases in locomotive and play behaviors and a progressive decline of social clinging activity throughout the course of the study. These behavioral trends are characteristic of "normal" social development of rhesus monkeys reared in social groups (Hansen, 1966; Harlow & Harlow, 1965; Rosenblum, 1961; Suomi & Harlow, 1971). In contrast, both separated groups showed little, if any, increase in locomotion or development of play activity—clear evidence of maturational arrest.

It may be noted that among human infants who exhibited anaclitic depression following maternal separation, common symptoms included "retardation of development" and "retardation of reaction to stimuli" (Spitz, 1946, p. 316). If one is willing to equate maturational arrest in monkey infants with retardation of development in human infants, then the data from the present study add support to the notion that infant separation of monkeys may serve as a model of human anaclitic depression.

The results of the present study also indicated that the nature of the subjects' environment during the separation periods had a significant effect on their subsequent activity.

TABLE 2

RESULTS OF ANOVA, SECOND SEPARATION SERIES (SEPARATIONS 13–20)

Category	*Group* (G)	*Time* (T)	G × T
Self-mouth	8.45*	<1.	1.55
Environmental exploration	<1	<1	<1
Self-clasp	8.46*	2.53	1.73
Locomotion	<1	<1	<1
Partner cling	3.70	1.03	<1
Play	4.32	<1	1.18

Note. Tabled numbers are *F* scores, $df = 1/6$ in all cases. No Duncan tests were performed on these data.
* $p < .05$.

TABLE 3

RESULTS OF ANOVA AND DUNCAN TESTS, BASE-LINE PERIODS

Category	Group (G)[a]	Time block (T)[b]	G × T[b]	Duncan test
Self-mouth	1.85	1.54	3.47*	VS3 > VS1, CS3, C1* VS3 > CS1, CS2, C2, C3** VS2 > CS1, CS2, C1**
Environmental exploration	1.10	5.22**	1.93	3 < 1,2*
Self-clasp	8.54**	16.60***	22.46***	VS3 > VS1, VS2, CS1, CS2, CS3, C1, C2, C3**
Locomotion	9.85**	<1	13.45	C3 > VS1,VS2, VS3, CS1, CS2, CS3, C1, C2** C2 > VS3, CS3** VS1 > VS3, CS3** VS2 > VS3, CS3* CS2 > VS3, CS3*
Partner cling	23.10***	24.47***	11.09**	C3 < VS1, VS2, VS3, CS1, CS2, CS3, C1** C2 < VS1, VS3, CS1, CS2, CS3, C1** VS2 < VS1, VS3, CS1, CS2, CS3, C1**
Play	9.12**	5.91**	6.84**	C3 > VS1, VS2, VS3, CS1, CS2, CS3, C1, C2**

Note.—Tabled numbers are *F* scores. Abbreviations for Duncan tests: VS = vertical chamber—separated, CS = cage—separated; 1 = 3-month base line, 2 = 6-month base line, 3 = 9-month base line.

[a] $df = 1/6$.

[b] $df = 2/12$.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

Although both the chamber-separated subjects in the present study and the cage-separated subjects in the Suomi et al. (1970) study exhibited protest–despair reactions to the separations, their behaviors during the reunion periods and the latter base-line periods differed significantly. Specifically, the chamber-separated subjects exhibited higher levels of self-directed activity (self-mouth, self-clasp) and lower levels of social contact activity (partner cling) than their cage-separated counterparts. In previous studies involving vertical chamber confinement of young monkeys (Harlow & Suomi, 1971; Suomi & Harlow, 1972), it was found that a consistent and dramatic elevation of self-clasping behavior followed subject incarceration. Further, when chambered subjects were permitted the opportunity for social interaction they showed little, if any, clinging behavior directed toward other monkeys. Thus, the direction of behavioral change of the chambered subjects in the present study appears consistent with that previously found to follow vertical chamber confinement. Whether these changes can be traced specifically to variables such as chamber shape, chamber size, or more likely, to a combination of these and other variables is a subject of ongoing researches at our laboratory.

The significance of the differences between the chamber-separated and cage-separated subjects' reunion and base-line behaviors is open to considerable speculation. One possible interpretation is that the patterns shown by the chamber-separated subjects represent the monkey equivalent of "detachment" behaviors shown by some human children following maternal reunion which Bowlby (1960) described. Certain primate investigators have reported isolated incidents of detachmentlike activity in infant monkeys reunited with their mothers following separation (Abrams, 1969; Hinde & Spencer-Booth, 1971; Kaufman & Rosenblum, 1967b; Seay et al., 1962). However, these incidents did not involve all separated infants in each study but rather specific individuals. Moreover, the detachmentlike behaviors were generally of short duration, usually disappearing within a few days after reunion. In contrast, all chamber-separated monkeys in the present study exhibited excessive self-clasping and a paucity of partner clinging throughout most of the 6-month

period of investigation. The equation of hostility and/or indifference in human children with increased self-clasping and decreased partner clinging in monkey infants may perhaps be overly anthropomorphic. At any rate, additional data clearly are needed before definite conclusions can be drawn regarding the occurrence of detachment as a consistent and predictable component of monkey reaction to social separation involving vertical chamber confinement.

Finally, it may be pointed out that while the chamber-separated subjects in the present study exhibited levels of most behaviors significantly different from those of nonseparated control monkeys, their behaviors following incarceration did not appear to be as severely psychopathological as the behaviors of chamber-confined subjects in previous studies (Harlow & Suomi, 1971; Suomi & Harlow, 1972). Specifically, the present subjects did not exhibit the debilitating levels of self-clasping and huddling or the almost total absence of social activity of any form that characterized the behavioral repertoire of the monkeys in the previous studies. These discrepancies in reaction to vertical chamber confinement may have resulted from any number of factors, including differences between studies in age of subjects during confinements, number of confinements, and duration of confinements. However, we believe that the primary variable responsible for the observed differences in severity of reaction to vertical chamber confinement lay in the nature of subject social experience prior to confinement. Specifically, the monkeys in the present study were reared as a social group and had the opportunity to form social bonds prior to incarceration, whereas monkeys in the previous studies were individually housed and thus had no opportunity to form social bonds prior to confinement. The present author believes that this fact is important for two reasons.

First, a wealth of data has indicated that monkeys housed individually from birth develop behavior repertoires vastly different from those of monkeys reared in a more socially enriched environment (for a review of the pertinent literature, see Harlow & Harlow, 1969). Thus, one would not expect the subjects in the present and previous studies to be behaviorally identical prior to the manipulation of vertical chamber confinement, and therefore, it should not be surprising that the absolute levels of their behaviors were not identical following the manipulation although the direction of behavioral changes following incarceration was similar.

Second, it is argued that chamber confinement per se represented a qualitatively different manipulation for the subjects in the studies in question. Gewirtz (1961), in a discussion of experiential factors affecting human infant behavior, makes a theoretical distinction between *privation,* that is, the denial of the opportunity to form social bonds, and *deprivation,* that is, the interruption of an existing social bond. He argues that these two conditions have different behavioral consequences. The present author believes that a similar distinction between chambered monkeys in previous researches and the chambered subjects in the present study is appropriate. The manipulation of chamber confinement for the former subjects represented an extention of an existing privation condition, while for the latter subjects it represented the imposition of a deprivation condition.

The present data thus suggest that prior social experiences should be taken into consideration when evaluating the effects of a manipulation such as repetitive chamber confinement on monkey subjects. The present author has steadfastly maintained that complex behaviors exhibited by organisms as complex as any primate are unlikely to be mediated by a single variable. The present data are consistent with this position.

REFERENCES

ABRAMS, P. S. Age and the effects of separation on mother and infant rhesus monkeys (*Macaca mulatta*). *Annual Meeting of the American Association of Physical Anthropologists,* Mexico City, 1969.

BLOMQUIST, A. J., & HARLOW, H. F. The infant rhesus monkey program at the University of Wisconsin Primate Laboratory. *Proceedings of the Animal Care Panel,* 1961, 11, 57–64.

BOELKINS, R. C. The development of social behavior in the infant rhesus monkey following a period of social isolation. Unpublished master's thesis, University of Wisconsin, 1963.

BOWLBY, J. Grief and mourning in infancy and early childhood. *Psychoanalytic Study of the Child,* 1960, **15**, 9–52.

BOWLBY, J. *Attachment and loss,* Vol. 1. New York: Basic Books, 1969.

CLARK, D. L. Immediate and delayed effects of early, intermediate, and late isolation in the rhesus monkey. Unpublished doctoral dissertation, University of Wisconsin, 1968.

DUNCAN, D. B. Multiple range and multiple *F* tests. *Biometrics,* 1955, **11**, 1–42.

GEWIRTZ, J. L. A learning analysis of the effects of normal stimulation, privation, and deprivation on the acquisition of social motivation and attachment. In B. M. Foss (Ed.), *Determinants of infant behavior.* Vol. 1. London: Methuen, 1961.

GRIFFIN, G. A., & HARLOW, H. F. Effects of three months of total deprivation on social adjustment and learning in the rhesus monkey. *Child Development,* 1966, **37**, 533–547.

HANSEN, E. W. The development of maternal and infant behavior in the rhesus monkey. *Behaviour,* 1966, **27**, 107–149.

HARLOW, H. F., & HARLOW, M. K. The affectional systems. In A. M. Schrier, H. F. Harlow, & F. Stollnitz (Eds.), *Behavior of nonhuman primates,* Vol. 2. New York: Academic Press, 1965.

HARLOW, H. F., & HARLOW, M. K. Effects of various mother-infant relationships on rhesus monkey behaviors. In B. M. Foss (Ed.), *Determinants of infant behavior.* Vol. 4. London: Methuen, 1969.

HARLOW, H. F., & SUOMI, S. J. Production of depressive behaviors in young monkeys. *Journal of Child Autism and Schizophrenia,* 1971, **1**, 246–255.

HARLOW, H. F., SUOMI, S. J., & MCKINNEY, W. T. Experimental production of depression in monkeys. *Mainly Monkeys,* 1970, **1**, 6–12.

HINDE, R. A., & SPENCER-BOOTH, Y. Effects of brief separations from mother on rhesus monkeys. *Science,* 1971, **173**, 111–118.

HINDE, R. A., SPENCER-BOOTH, Y., & BRUCE, M. Effects of 6-day maternal deprivation on rhesus monkey infants. *Nature,* 1966, **210**, 1021–1033.

JENSEN, G. D., & TOLEMAN, C. W. Mother-infant relationship in the monkey. *Macaca nemestrina:* The effect of brief separation and mother-infant specificity. *Journal of Comparative and Physiological Psychology,* 1962, **55**, 131–136.

KAUFMAN, I. C., & ROSENBLUM, L. A. Depression in infant monkeys separated from their mothers. *Science,* 1967, **155**, 1030–1031. (a)

KAUFMAN, I. C., & ROSENBLUM, L. A. The reaction to separation in infant monkeys: Anaclitic depression and conservation-withdrawal. *Psychosomatic Medicine,* 1967, **29**, 648–675. (b)

LIDDELL, H. S., JAMES, W. T., & ANDERSON, O. D. The comparative physiology of the conditioned motor reflexes based on experiments with the pig, dog, sheep, goat, and rabbit. *Comparative Physiology Monographs,* 1934, **11**, 1–89.

MASSERMAN, J. L. *Behavior and neurosis.* Chicago: University of Chicago Press, 1943.

MCKINNEY, W. T., & BUNNEY, W. E. Animal model of depression. *Archives of General Psychiatry,* 1969, **21**, 240–248.

PAVLOV, I. C. *Conditioned reflexes.* London: Oxford University Press, 1927.

ROSENBLUM, L. A. The development of social behavior in the rhesus monkey. Unpublished doctoral dissertation, University of Wisconsin, 1961.

ROWLAND, G. L. The effects of total social isolation upon learning and social behavior in rhesus monkeys. Unpublished doctoral dissertation, Universtiy of Wisconsin, 1964.

SCHMALE, A. The role of depression in health and disease. Paper presented at the 137th annual convention of the American Association for the Advancement of Science, Chicago, 1970.

SEAY, B., HANSEN, E. W., & HARLOW, H. F. Mother-infant separation in monkeys. *Journal of Child Psychology and Psychiatry,* 1962, **3**, 123–132.

SEAY, B., & HARLOW, H. F. Maternal separation in the rhesus monkey. *Journal of Nervous and Mental Diseases,* 1965, **140**, 434–440.

SPITZ, R. A. Anaclitic depression. *Psychoanalytic Study of the Child,* 1946, **2**, 313–347.

SUOMI, S. J., & HARLOW, H. F. Apparatus conceptualization for psychopathological research in monkeys. *Behavior Research Methods and Instrumentation,* 1969, **1**, 247–250.

SUOMI, S. J., & HARLOW, H. F. Monkeys at play. *Natural History,* 1971, **80**, 72–76.

SUOMI, S. J., & HARLOW, H. F. Depressive behavior in young monkeys subjected to vertical chamber confinement. *Journal of Comparative and Physiological Psychology,* 1972, **180**, 11–18.

SUOMI, S. J., HARLOW, H. F., & DOMEK, C. J. Effect of repetitive infant-infant separation of young monkeys. *Journal of Abnormal Psychology,* 1970, **76**, 161–172.

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